SEMINAR III · WEEK 6 · FOUNDATIONS

Ratios, Rates & Proportions

This week's big question: what two quantities am I comparing?

SESSION A

Ratios and Rates

WHAT DOES A RATIO COMPARE?

Part-to-part, or part-to-whole?

12-player team

Part-to-part: forwards to defenders = **5:7**

Part-to-whole: forwards to the whole team = **5/12** — not 5/7

WRITING AND SIMPLIFYING RATIOS

The same ratio can be written three ways: 3 to 2, 3:2, or 3/2. Order matters — 3:2 and 2:3 describe different comparisons.

> Simplify 18:12 using the greatest common factor (6): $18 \div 6 = 3$, $12 \div 6 = 2$ → **3:2**. Same relationship, smaller numbers.

EQUIVALENT RATIOS

Scale up or down by multiplying — not adding.

> If a recipe ratio needs more sugar, don't add the same amount to every ingredient. Multiply or divide *every* quantity by the same factor.

WHAT IS A RATE?

A rate is a ratio that compares quantities with *different* units — miles to hours, dollars to items. Every rate is a ratio, but not every ratio is a rate. Keep the units attached; they're part of the answer.

UNIT RATES

Divide so the second quantity becomes exactly 1.

> $104 for 8 hours of work: 104 ÷ 8 = **$13 per hour**.

A unit rate isn't automatically "bigger number ÷ smaller number" — it's whichever quantity the question calls the reference, the "per one" unit.

COMPARING RATES: BEST BUY

Calculate each unit rate, then compare directly.

> A higher total price doesn't automatically mean a worse deal — the unit rate decides value, not the sticker price.

BEFORE WE MOVE ON

1. Generate an equivalent ratio
2. Calculate a unit rate
3. Pick the better value between two rates

SESSION B

Proportions and Unit Conversion

SOLVING A PROPORTION

4/5 = x/30

$$\frac{4}{5} = \frac{x}{30}$$

Cross multiply: 4×30 = 5×x, so 120 = 5x, **x = 24**. This only works when the proportion is set up with matching quantities in matching positions.

UNIT CONVERSION

A conversion factor is just a ratio, arranged so the unit you don't want cancels out. Flip it, and it cancels the wrong thing — watch for a result that's obviously the wrong size, and let estimation catch it.

DOES MY ANSWER MAKE SENSE?

Check	Ask
Size	Is the answer approximately the size I expected?
Direction	Did I scale up or down the way the situation requires?
Units	Did the units cancel or combine the way they should?
Context	Does this answer make sense in the real situation?

BEFORE WE MOVE ON

1. Set up a proportion from a word problem
2. Solve it
3. Check your solution

Then: mixed practice, error analysis, Week 6 check, reflection.

TRANSITION TO WEEK 7

Variables & Expressions

A proportion asks you to find an unknown quantity. Algebra gives you a more general way to represent and solve for unknowns.